

Objectives

- Explain branching and merging
- Explain about creating a branch request in GitLab
- Explain about creating a merge request in GitLab

In this hands-on lab, you will learn how to:

- Construct a branch, do some changes in the branch, and merge it with master (or trunk)

Prerequisites

The following are the pre-requisites to complete this hands-on lab:

- Setting up Git environment with P4Merge tool for Windows

Notes*:

Please follow the below steps for creating a free account in GitHub.
Do not use cognizant credentials to login to GitHub.

Estimated time to complete this lab: **30 minutes**.

Please follow the instruction to complete the hands-on. Each instruction expects a command for the Git Bash.

Branching:

1. Create a new branch “**GitNewBranch**”.
2. List all the local and remote branches available in the current trunk. Observe the “*” mark which denote the current pointing branch.
3. Switch to the newly created branch. Add some files to it with some contents.
4. Commit the changes to the branch.
5. Check the status with “**git status**” command.

Merging:

1. Switch to the master
2. List out all the differences between trunk and branch. These provide the differences in command line interface.
3. List out all the visual differences between master and branch using **P4Merge tool**.
4. Merge the source branch to the trunk.
5. Observe the logging after merging using “**git log --oneline --graph --decorate**”
6. Delete the branch after merging with the trunk and observe the git status.

SOLUTION

```
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT> cd WEEK-6
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6> cd 03.GIT
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\03.GIT> git init

Initialized empty Git repository in C:/Users/valla/OneDrive/Desktop/DN-JAVA-FSE-REACT/WEEK-6/03.GIT/.git/
```

```
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\03.GIT> echo Hello Git > welcome.txt
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\03.GIT> git add .
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\03.GIT> git commit -m "Initial commit"
[master (root-commit) 08a3352] Initial commit
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 welcome.txt
```

```
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\03.GIT> git branch GitNewBranch
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\03.GIT> git branch
  GitNewBranch
* master
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\03.GIT> git checkout GitNewBranch
Switched to branch 'GitNewBranch'
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\03.GIT> git branch
  GitNewBranch
* master
```

```
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\03.GIT> echo This is branch work > branch.txt
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\03.GIT> git add .
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\03.GIT> git commit -m "Added branch.txt"
[GitNewBranch d332673] Added branch.txt
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 branch.txt
```

```
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\03.GIT> git status
On branch GitNewBranch
nothing to commit, working tree clean
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\03.GIT> git checkout master
Switched to branch 'master'
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\03.GIT> git diff master GitNewBranch
diff --git a/branch.txt b/branch.txt
new file mode 100644
index 0000000..0ab8b4f
Binary files /dev/null and b/branch.txt differ
```

```
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\03.GIT> git merge GitNewBranch
Updating 08a3352..d332673
Fast-forward
 branch.txt | Bin 0 -> 50 bytes
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 branch.txt
```

```
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\03.GIT> git log --oneline --graph --decorate
* d332673 (HEAD -> master, GitNewBranch) Added branch.txt
* 08a3352 Initial commit
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\03.GIT> git branch -d GitNewBranch
error: branch 'GitNewBranch' not found
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\03.GIT> git branch -d GitNewBranch
Deleted branch GitNewBranch (was d332673).
PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\03.GIT> git branch
* master
```